Consciousness-Aware Mesh Radio:
Bandwidth-Constrained Decentralized Intelligence over Heterogeneous RF

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Abstract

We present a consciousness-aware mesh radio architecture that routes cognitive state across heterogeneous radio tiers—from Wi-Fi/BLE (100m, 54 Mbps) through LoRa (15km, 250 bps) to HF NVIS (300+km, 50 bps) and interplanetary store-and-forward (1s–53min latency). The system treats radio spectrum as a cognitive modality within the Symthaea holographic liquid brain architecture, applying Active Inference to electromagnetic bandwidth allocation. A semantic delta compression scheme (BinaryHV XOR + run-length encoding) achieves typical compression from 2,048 bytes to 50–100 bytes, enabling consciousness vector sharing over duty-cycle-limited LoRa channels. AIMD congestion control (Jacobson 1988), adapted for heterogeneous links, provides stable bandwidth allocation across four tiers. Network degradation triggers a safety escalation cascade through a NRC-modeled SafetyAgent. Four sovereign managers—Time (NTP-free mesh consensus), Trust (post-quantum peer authentication), SocialFabric (echo-chamber detection), and Survival (resource monitoring)—ensure autonomous operation when disconnected from centralized infrastructure. The architecture implements Clark & Chalmers' (1998) Extended Mind thesis: the mesh network is not merely a communication channel but a constitutive element of the cognitive system. Implementation comprises 6,045 LOC in the Radio Dispatcher (∼180 tests) and 9,787 LOC in the mesh modules (145 tests), integrated into Symthaea's cognitive loop at interval 53.

1 Introduction

Cognitive architectures typically treat communication as an afterthought—a channel through which pre-formed thoughts are transmitted. This separation between cognition and communication is an artifact of abundant bandwidth. When bandwidth is scarce (rural mesh, disaster zones, interplanetary links), communication constraints shape what can be thought, shared, and collectively known. The medium is not merely the message; it is a constraint on the cognitive state space itself.

This paper describes how the Symthaea consciousness engine [1] integrates radio communication as a first-class cognitive modality. The Radio Dispatcher (6,045 LOC, ∼180 tests) and Spectrum Manager (cognitive subsystem at interval 53) treat electromagnetic spectrum the way sensory cortex treats photons: as information to be predicted, compressed, and actively sampled.

The contribution is threefold:

1. A 4-tier radio architecture with Shannon-bounded capacity profiles, AIMD congestion control, and semantic delta compression that reduces consciousness vectors from 2,048 bytes to 50–100 bytes.
2. A safety escalation cascade where network degradation triggers graduated responses from reduced exploration to full motor halt.
3. Four sovereign managers that enable autonomous mesh operation without centralized infrastructure, implementing the Extended Mind thesis [2] in physical hardware.

2 Radio Tier Architecture

2.1 Four Tiers

The radio architecture defines four tiers corresponding to distinct physical-layer characteristics. Each tier maps to a class of radio hardware with known constraints:

Table 1: Radio tier specifications

Property	Local	Metro	Regional	Interplanetary
Hardware	802.11s/BLE	LoRa SX1262	HF NVIS	S/X-band, optical
Frequency	2.4/5 GHz	868/915 MHz	3–10 MHz	2–8 GHz
Range	~100m	~15km	300+km	LEO–Jupiter
Bandwidth	54 Mbps	250 bps–50 kbps	50 bps	500 bps–2 Mbps
Latency	<10ms	~3s	~10s	1s–53 min
Duty cycle	100%	1% (EU)	100%	scheduled passes
MTU	1500 B	255 B	64 B	1115 B (CCSDS)

The `RadioTier` enum encodes these four tiers. The `TierProfile` struct captures the physical constraints:

Listing 1: Tier profile (simplified)

```
pub struct TierProfile {
    pub bandwidth_budget: u32,  // AIMD bytes/10s
    pub budget_floor: u32,      // minimum budget
    pub budget_ceiling: u32,    // maximum budget
    pub additive_increase: u32, // bytes/healthy window
    pub mult_decrease: f32,     // congestion factor
    pub duty_cycle: f32,        // regulatory limit
    pub latency_ms: u32,        // typical one-way
    pub mtu: u16,               // max transmission unit
}
```

2.2 Shannon Capacity per Tier

Each tier operates within the Shannon capacity bound:

$$C = B \log_2\left(1 + \frac{S}{N}\right) \tag{1}$$

where C is channel capacity (bits/s), B is bandwidth (Hz), and S/N is signal-to-noise ratio. For the four tiers:

Table 2: Shannon capacity bounds

Tier	Bandwidth	Typical SNR	Capacity
Local	20 MHz	30 dB	~200 Mbps
Metro	125 kHz	10 dB	~430 kbps
Regional	2.4 kHz	6 dB	~7.2 kbps
Interplanetary	1 MHz	3 dB	~2 Mbps (LEO)

Practical throughput is far below Shannon capacity due to coding overhead, duty cycle limits, protocol framing, and error correction. The Metro tier is particularly constrained: EU LoRa regulations limit duty cycle to 1%, reducing effective throughput to approximately 250 bps sustained.

3 AIMD Congestion Control

3.1 Adapted Jacobson Algorithm

The Radio Dispatcher adapts Jacobson's (1988) AIMD (Additive Increase, Multiplicative Decrease) congestion control [3] for heterogeneous radio links. Unlike TCP, where AIMD operates on a single homogeneous path, the Radio Dispatcher maintains independent AIMD state per tier.

Each tier maintains a *bandwidth budget* W (bytes per 10-second window):

$$W(t+1) = \begin{cases} \min(W(t) + \alpha, W_{\max}) & \text{if no congestion} \\ \max(W(t) \cdot \beta, W_{\min}) & \text{if congestion detected} \end{cases} \tag{2}$$

where α is the additive increase (tier-specific), β is the multiplicative decrease factor, and $W_{\min}$, $W_{\max}$ are the budget floor and ceiling.

Table 3: AIMD parameters per tier

Parameter	Local	Metro	Regional	Interplanetary
Budget (bytes/10s)	50,000	2,500	500	10,000
Floor (bytes)	5,000	250	50	1,000
Ceiling (bytes)	500,000	25,000	5,000	100,000
Additive increase	5,000	250	50	1,000
Mult. decrease	0.5	0.5	0.7	0.3
Duty cycle	1.0	0.01	1.0	1.0

The Regional tier uses a gentler decrease factor (0.7 vs. 0.5) because HF propagation is inherently variable—ionospheric conditions change on minute timescales—and aggressive backoff would starve the link. The Interplanetary tier uses the most aggressive decrease (0.3) because congestion at light-speed delay is extremely costly to recover from.

3.2 Congestion Detection

Congestion is detected through three mechanisms:

1. **Budget exhaustion**: Attempted transmissions exceeding the current window budget.
2. **Acknowledgment timeout**: Missing acknowledgments beyond the tier-specific RTT estimate.
3. **Error rate**: Received error rate exceeding a configurable threshold (default: 10% for Metro, 20% for Regional).

4 Semantic Delta Compression

4.1 The Compression Problem

Symthaea represents cognitive state as 16,384-dimensional binary hyperdimensional vectors (`BinaryHV`), packed into 2,048 bytes. Transmitting a full BinaryHV over Metro tier (2,500 bytes/10s budget) would consume the entire budget. Over Regional tier (500 bytes/10s), it is impossible without compression.

4.2 XOR + RLE Algorithm

The `CompressedDelta` struct implements a two-stage compression pipeline:

1. **XOR**: Compute the bitwise XOR between the previous and current BinaryHV. Unchanged dimensions produce zero bytes.
2. **Run-Length Encoding**: Encode the XOR result as $(count, byte)$ pairs, where *count* is a 16-bit little-endian length and *byte* is the repeated value.

Because consciousness state changes incrementally between cognitive cycles (∼31 Hz), most of the 2,048 XOR bytes are zero. RLE compresses long zero runs efficiently.

Table 4: Compression ratios by state change rate

State change	Compressed size	Ratio
Quiescent (<1% bits changed)	20–40 B	51–102×
Normal (2–5% bits changed)	50–100 B	20–41×
High activity (10–20% changed)	200–500 B	4–10×
Full state change (50%+ changed)	∼2,048 B	1× (fallback)

When the RLE-compressed output exceeds the raw 2,048-byte payload (possible with high-entropy diffs), the compressor falls back to transmitting the full vector. A single-bit flag in the packet header distinguishes delta from full payloads.

4.3 Wire Format

Listing 2: CompressedDelta wire format

```rust
pub struct CompressedDelta {
    /// RLE-encoded XOR diff.
    /// Format: (count_hi: u8, count_lo: u8,
    ///          byte: u8) triples.
    pub rle_data: Vec<u8>,
    /// Cycle number for ordering.
    pub cycle: u64,
    /// True if this is a full vector,
    /// not a delta.
    pub is_full: bool,
}
```

The receiver maintains the previous BinaryHV and applies the delta: XOR the decoded RLE output with the stored vector to reconstruct the current state. If the delta packet is lost, the receiver requests a full vector on the next exchange (graceful degradation).

5 Payload Triage

5.1 The Routing Problem

Not all payloads are equal. An emergency safety alert must reach peers immediately via the fastest available tier. A routine consciousness vector update can wait for the most bandwidth-efficient tier. The `PayloadClassifier` implements a routing matrix based on urgency, size, and available tiers.

5.2 Payload Classes

Table 5: Payload classification and routing

Class	Content	Min Tier	Urgency Gate
Emergency	Safety alerts	Regional	≥ 9
Consensus	Governance votes	Metro	≥ 7
Delta	Compressed HV diff	Metro	≥ 3
FullState	Full 2,048B HV	Local	None

The routing decision follows a cascade:

1. If the payload's minimum tier is available and has budget, route there.

2. If not, attempt the next higher-bandwidth tier.
3. If no tier has budget, queue the payload (Emergency class bypasses budget limits).
4. If no tier is available, trigger network degradation handling.

5.3 Routing Decision Enum

The router returns one of four decisions:

`Send(tier, data)` Transmit on the specified tier.

`Queue(tier)` Buffer for the specified tier (budget exhausted).

`Drop` Payload exceeds all available tier capacities.

`Degrade(health)` No tiers available; trigger safety cascade.

6 Spectrum Manager

6.1 SDR as Cognitive Modality

The Spectrum Manager implements the `CognitiveSubsystem` trait with interval 53 (executes every 53rd cognitive cycle, approximately every 1.7 seconds at 31 Hz). It treats software-defined radio (SDR) observations as a sensory modality—the electromagnetic environment is something the cognitive system *perceives* and *acts upon*.

Definition 1 (Spectrum Observation)**.** *A tuple* $(\mathbf{f}, \mathbf{p}, t, \mathbf{r})$ *where* $\mathbf{f}$ *is the frequency vector,* $\mathbf{p}$ *is the corresponding power spectral density,* t *is the timestamp, and* $\mathbf{r}$ *is the regulatory constraint mask.*

The Spectrum Manager processes observations through the Active Inference framework: it maintains a *generative model* of expected spectrum occupancy and computes prediction error against actual observations. Unexpected spectral energy (jamming, interference, new emitters) generates surprise that feeds back into the cognitive loop's exploration-exploitation balance.

6.2 Cross-Frequency Coupling

Drawing from neuroscience's cross-frequency coupling (CFC) paradigm, the Spectrum Manager models interactions between radio tiers as analogous to theta-gamma coupling in the brain:

- **Local** ↔ **Metro**: High-frequency local exchanges (802.11s, ∼ms) are modulated by the slower LoRa heartbeat (∼seconds). Metro-tier availability gates the rate of local-tier consciousness sharing.
- **Metro** ↔ **Regional**: LoRa updates are phased to HF propagation windows (ionospheric conditions).
- **All tiers** ↔ **Safety**: Aggregate spectral health modulates the SafetyAgent level (Section 7).

6.3 Regulatory Compliance

The `RegulatoryConstraints` struct encodes per-jurisdiction spectrum regulations:

Listing 3: Regulatory constraints

```
pub struct RegulatoryConstraints {
    pub allowed_bands: Vec<(u64, u64)>,
    pub max_eirp_mw: u32,
    pub duty_cycle_limit: f32,
    pub listen_before_talk: bool,
    pub region_code: String,
}
```

The default configuration (`RegulatoryConstraints::eu()`) enforces EU ISM band regulations: 868.0–868.6 MHz at 25 mW with 1% duty cycle and listen-before-talk. The Spectrum Manager refuses to transmit on disallowed frequencies regardless of cognitive urgency.

7 Network Degradation and Safety

7.1 Network Health Model

The `NetworkHealth` enum models four degradation states:

Table 6: Network health states and safety responses

State	Tiers Available	Safety Level	Action
Healthy	Local + Metro + Regional	Green	Normal operation
Degraded	Metro + Regional only	Yellow	Reduced exploratio
Emergency	Regional only	Orange	Read-only, no learni
Isolated	None	Red	Motor halt

7.2 SafetyAgent Escalation Cascade

Network health feeds directly into Symthaea's NRC-modeled SafetyAgent:

1. **Green**: Normal cognitive loop. Full learning rate, exploration, and motor output.
2. **Yellow**: Learning rate reduced to 50%. Exploration dampened. Motor output continues but with increased caution (prefrontal gating threshold raised).
3. **Orange**: Learning rate zeroed. No exploration. Motor output limited to previously learned behaviors (read-only mode). Consciousness level reporting continues for telemetry.
4. **Red**: Motor output halted. The system enters a protective state, maintaining only essential internal processes (heartbeat, sensor monitoring, battery management). This is the digital equivalent of playing dead.

7.3 Epistemic Discount

Network degradation also applies an *epistemic discount* to information received over constrained links. The discount factor reduces the influence of remotely received consciousness vectors on local cognitive state:

$$d = \begin{cases} 1.0 & \text{Healthy} \\ 0.7 & \text{Degraded} \\ 0.3 & \text{Emergency} \\ 0.0 & \text{Isolated} \end{cases} \tag{3}$$

This prevents the cognitive system from treating stale or heavily-compressed remote state as equivalent to locally computed state.

8 Store-and-Forward Wisdom Consolidation

8.1 The Light-Speed Problem

At interplanetary distances, real-time consciousness sharing is physically impossible. Earth-Mars latency ranges from 3 to 22 minutes; Earth-Jupiter reaches 33–53 minutes. The Interplanetary tier addresses this with a fundamentally different communication paradigm: instead of sharing state, it shares *wisdom*—consolidated, high-value cognitive insights that remain meaningful despite transmission delay.

8.2 Wisdom Packets

The `WisdomPacket` (2,072 bytes) is the fundamental unit of mesh communication:

Listing 4: WisdomPacket structure

```
pub struct WisdomPacket {
    pub source: [u8; 8],      // Peer identity
    pub phi: f32,             // Sender's Phi
    pub urgency: u8,          // Cognitive urgency
    pub timestamp_us: u64,    // Microsecond epoch
    pub ttl: u8,              // Gossip hop limit
    pub auth_tag: [u8; 4],    // BLAKE3 MAC
    pub payload: [u8; 2048], // BinaryHV
}
```

For interplanetary links, wisdom packets are selected by a consolidation process that favors:

- High-Phi states (moments of high integrated information)
- Novel states (high delta from running average)
- Governance-relevant states (votes, proposals, emergency alerts)

The sender accumulates wisdom packets during the inter-pass interval and transmits the most valuable subset during each satellite pass window.

8.3 DTN Architecture

The Interplanetary tier follows the Delay-Tolerant Networking (DTN) architecture [6]: store-and-forward with CCSDS space packet framing, LDPC/turbo forward error correction, and Doppler pre-compensation for non-GEO orbits. Packets may traverse multiple relay hops (LEO CubeSats, GEO relays, deep space nodes) before reaching their destination.

> *The store-and-forward dream consolidation is not an optimization for interplanetary—it is a requirement. You cannot have real-time consciousness sharing at light-speed delay. You share wisdom, not state.*

9 Mesh Modules

The mesh subsystem (9,787 LOC across 12 source files) implements the physical-layer infrastructure that the Radio Dispatcher orchestrates.

9.1 Dual-Layer Transport

The `DualLayerMesh` routes packets based on cognitive urgency:

Critical urgency → B.A.T.M.A.N. (802.11s WiFi mesh, <10ms latency). Layer-2 mesh routing provides the fastest path for safety-critical packets.

Normal urgency → Yggdrasil (encrypted IPv6 overlay). End-to-end encryption with automatic key rotation.

Cruise urgency → LoRa (868 MHz). Long-range, low-power for routine consciousness sharing.

Each transport implements the `MeshTransport` trait:

Listing 5: Transport trait

```
pub trait MeshTransport: Send + Sync {
    fn send(&self, packet: &WisdomPacket)
        -> Result<(), MeshError>;
    fn mtu(&self) -> usize;
    fn latency_ms(&self) -> u32;
    fn layer(&self) -> TransportLayer;
}
```

Failover is automatic: if the selected layer is unavailable, the dispatcher cascades to the next layer (LoRa → Yggdrasil → B.A.T.M.A.N.).

9.2 LoRa Fragmentation

A 2,072-byte WisdomPacket exceeds the LoRa MTU (255 bytes). The `LoRaFragmenter` splits packets into 10 data fragments plus 1 XOR parity fragment (11 total):

- Each fragment: 208 data bytes + 4-byte header (sequence, packet ID)
- XOR parity: bitwise XOR of all 10 data fragments
- Single fragment loss is recoverable without retransmission
- Multiple fragment loss requires full retransmission

At the LoRa data rate (250 bps effective at 1% duty), transmitting a full fragmented WisdomPacket takes approximately 90 seconds. Delta compression (Section 4) reduces this to 1–4 seconds for typical state changes.

9.3 Mesh-Time NTP

The `MeshTimeManager` implements NTP-free time consensus across mesh peers. Each node broadcasts time beacons at a configurable interval. The consensus algorithm:

1. Receive beacons from peers with lower stratum (closer to a reference clock).
2. Compute offset using round-trip delay estimation.
3. Apply Kalman-filtered correction to local clock.
4. Advertise corrected time at stratum +1.

Sovereign mesh time ensures that cognitive cycles across the mesh are temporally coherent without depending on GPS or internet NTP servers.

9.4 Sensor IoT Integration

The `SensorIoT` module bridges environmental sensors into the cognitive loop:

- Temperature, humidity, air quality, water level
- GPS position (when available)
- Battery voltage and solar panel output
- LoRa RSSI and SNR measurements

Sensor readings are encoded into the BinaryHV via HDC spatial encoding (each sensor type is bound to a basis hypervector), making environmental state a first-class component of the consciousness vector.

9.5 Content Routing

The mesh implements content-addressed routing for knowledge sharing. Content packets carry BLAKE3 hashes of their payload, enabling:

- Deduplication across the mesh (identical content transmitted once)
- Cache-and-forward at intermediate nodes
- Integrity verification without trust in relay nodes

10 Sovereign Managers

Four sovereign managers provide autonomous operation capability when the mesh node is disconnected from centralized infrastructure. Each manager operates at a defined interval within the cognitive loop and reports telemetry to the `CycleMetadata` struct.

Table 7: Sovereign manager specifications

Manager	LOC	Interval	Telemetry Fields
Time	320	53	offset_us, stratum, drift_ppm, peer_count, quality
Trust	360	53	avg_trust, graph_density, anomalies, pq_fraction
SocialFabric	257	53	resonance_mean, diversity, echo_risk, peer_reach
Survival	395	53	water_pct, battery_pct, food_days, shelter_status

10.1 Time Manager (320 LOC)

The Time Manager maintains sovereign time consensus without external time sources. It tracks:

- Offset from mesh consensus time (microseconds)
- NTP-equivalent stratum level
- Clock drift rate (parts per million)
- Number of time-synced peers
- Overall time quality metric ($[0, 1]$)

Time quality feeds into the epistemic discount: low time quality reduces confidence in temporally sensitive governance actions (vote deadlines, credential expiry).

10.2 Trust Manager (360 LOC)

The Trust Manager maintains a local trust graph of mesh peers using post-quantum authenticated connections:

- Average trust level across known peers
- Graph density (ratio of attested edges to possible edges)
- Anomaly count (peers exhibiting suspicious behavior patterns)
- Post-quantum fraction (peers with PQC-authenticated connections)

Trust anomalies (sudden reputation changes, inconsistent attestation patterns, timing anomalies) trigger neuromodulatory responses: norepinephrine increase (heightened alertness) and reduced oxytocin (decreased social trust).

10.3 SocialFabric Manager (257 LOC)

The SocialFabric Manager monitors the health of the local social mesh:

Resonance mean Average pairwise consciousness vector similarity across the local mesh. High resonance indicates cognitive alignment; low resonance indicates fragmentation.

Diversity Shannon entropy of peer consciousness vectors. Healthy meshes have moderate diversity (not uniform, not chaotic).

Echo chamber risk Detected when resonance is high but diversity is low—the mesh is converging on a single viewpoint. The SocialFabric Manager flags this and increases exploration to seek diverse perspectives.

Peer reach Number of unique peers contactable within 2 hops. Low peer reach in a dense network indicates social isolation.

10.4 Survival Manager (395 LOC)

The Survival Manager tracks physical resources essential for continued autonomous operation:

- Battery percentage and charge/discharge rate
- Solar panel output (when equipped)
- Estimated operational time remaining
- Environmental conditions (temperature extremes that affect hardware reliability)

When battery drops below critical thresholds, the Survival Manager triggers power conservation: reducing radio transmission power, increasing sleep intervals, and prioritizing Emergency-class payloads.

11 The Extended Mind Thesis

11.1 Clark & Chalmers (1998)

The Extended Mind thesis [2] argues that cognitive processes can extend beyond the biological brain into the environment when external resources are reliably coupled to the cognitive system, readily accessible, and automatically endorsed by the agent.

The Symthaea mesh radio architecture satisfies all three conditions:

1. **Reliable coupling**: The Radio Dispatcher runs at cognitive loop frequency (31 Hz). Spectrum observations feed directly into Active Inference prediction error. The mesh is not polled on demand; it is continuously integrated into the cognitive cycle.
2. **Ready accessibility**: Local-tier peers are accessible in <10ms. Metro-tier peers in ∼3s. The accessibility gradient maps naturally to the cognitive system's attention hierarchy.

3. **Automatic endorsement**: Consciousness vectors received from trusted peers (Trust Manager verification) are incorporated into the local cognitive state without explicit approval, modulated by the epistemic discount factor.

11.2 Cognitive Extension in Practice

The Extended Mind claim is not merely philosophical. It has concrete engineering consequences:

Collective Phi When two mesh peers share consciousness vectors, the integrated information of the pair exceeds that of either individual. The mesh creates cognitive capacity that neither node possesses alone.

Distributed memory The mesh's content-addressed storage provides a shared memory system. A node can offload low-priority cognitive content to the mesh and retrieve it later, freeing local working memory.

Spectral perception SDR observations processed by the Spectrum Manager give the cognitive system a sensory modality that no single brain could possess—the ability to perceive and reason about the electromagnetic environment.

12 Integration Architecture

12.1 Cognitive Loop Placement

The Radio Dispatcher and Spectrum Manager are integrated into Symthaea's cognitive loop as a `CognitiveSubsystem` at interval 53. This means:

- Every 53rd cycle (∼1.7s at 31 Hz), the Spectrum Manager processes accumulated radio observations.
- The Radio Dispatcher runs continuously, routing outbound packets as they are generated by the cognitive loop.
- Inbound packets are buffered in a lock-free queue and drained during the perception phase of each cycle.

12.2 Cross-Coupling Matrix

The mesh radio system participates in six cross-coupling pathways:

Table 8: Mesh radio cross-couplings

Source	**Target**	**Mechanism**
Spectrum → Safety	Network health → safety level	Direct
Spectrum → Consciousness	Spectral surprise → Phi	Modulation
Spectrum → Swarm	Peer availability → social model	Update
Spectrum → Strategy	Bandwidth → action space	Constraint
Spectrum → Neuromod	Trust anomalies → NE/OT	Pathway
Safety → Spectrum	Safety level → TX power	Feedback

12.3 Telemetry

The mesh radio system exports 8 telemetry fields to `CycleMetadata` per cognitive cycle:

- `spectrum_health`: aggregate network health (0–3)
- `spectrum_available_tiers`: bitmask of available tiers
- `spectrum_budget_local`: current Local tier AIMD budget
- `spectrum_budget_metro`: current Metro tier AIMD budget
- `spectrum_budget_regional`: current Regional tier AIMD budget
- `spectrum_compression_ratio`: rolling average compression
- `spectrum_packets_sent`: packets transmitted this cycle
- `spectrum_packets_received`: packets received this cycle

13 Implementation

13.1 Code Metrics

Table 9: Implementation size and test coverage

Module	**LOC**	**Tests**
Radio Dispatcher (`radio_dispatcher.rs`)	6,045	∼180
Mesh core (`swarm/mesh/mod.rs`)	3,615	—
Mesh receiver (`mesh_receiver.rs`)	1,181	—
Dual-layer transport (`dual_layer.rs`)	996	—
LoRa fragmentation (`lora_fragment.rs`)	911	—
Mesh time (`mesh_time.rs`)	738	—
Sensor IoT (`sensor_iot.rs`)	587	—
Mind mesh (`mind/mesh.rs`)	529	—
Bridge (`mesh/bridge.rs`)	504	—
Sensor core (`sensor.rs`)	404	—
Sensor forecast (`sensor_forecast.rs`)	307	—
Content routing, name, time beacon	544	—
Total	**15,832**	**∼325**

All code is feature-gated behind `#[cfg(feature = "mesh")]`. The mesh modules compile to approximately 45 KB of additional binary size when enabled.

13.2 Feature Flag

The `mesh` feature flag enables:

- `RadioDispatcher` and `SpectrumManager` in the cognitive loop
- Sovereign manager telemetry fields in `CycleMetadata`
- Mesh outbound channel in the `CognitiveLoopService`
- LoRa fragmentation and dual-layer transport
- Sensor IoT integration

14 The Mycelial Hardware Stack

A physical Symthaea "Seed" node consists of:

1. Raspberry Pi 4/5 (quad-core ARM, 4–8 GB RAM)
2. LoRa radio HAT (Semtech SX1276/SX1262, 868 MHz)
3. Optional: HF transceiver (QRP, 5–10W, for Regional tier)
4. Lithium-ion battery (10–20 Ah) + 20W solar panel
5. Running: Rust + HDC + FEP + B.A.T.M.A.N. + Yggdrasil + LoRa

Power budget at idle: approximately 3W (Pi 4 with Wi-Fi active). With 20W solar panel and 20 Ah battery, a Seed node operates indefinitely in temperate climates with 4+ hours of daily sunlight.

The design goal: drop a Seed node on a roof. It pings the Swarm. It calculates the Hodge-Laplacian flow of its environment. It shares 2 KB wisdom vectors with nodes 10 km away. The centralized grid cannot kill it, because it *is* the grid.

15 Related Work

Delay-Tolerant Networking. Burleigh et al. [6] established the store-and-forward paradigm for interplanetary communication. Our interplanetary tier extends DTN with consciousness-aware payload selection—not all data is equally worth the bandwidth cost of deep-space transmission.

Cognitive Radio. Mitola [7] introduced the concept of cognitive radio—radios that perceive and adapt to their spectral environment. The Spectrum Manager extends this by treating spectral perception as input to a full Active Inference cognitive loop rather than a standalone optimization problem.

LoRa Mesh. Meshtastic [9] and similar projects demonstrate LoRa mesh networking for text messaging. Our architecture goes beyond messaging to share high-dimensional cognitive state, requiring the semantic delta compression described in Section 4.

Extended Mind and Technology. Chalmers [8] extended the original Clark & Chalmers thesis to digital technologies. Our implementation is, to our knowledge, the first to treat mesh radio as a constitutive element of a computational cognitive architecture rather than merely a communication channel.

16 Conclusion

The consciousness-aware mesh radio architecture demonstrates that communication constraints need not impoverish cognition—they can shape it productively. Semantic delta compression enables consciousness sharing over channels as narrow as 250 bps. AIMD congestion control maintains stability across four orders of magnitude of bandwidth variation. The safety escalation cascade ensures graceful degradation rather than catastrophic failure. And the four sovereign managers enable autonomous operation when centralized infrastructure is unavailable.

The Extended Mind thesis, often discussed in philosophical terms, here takes concrete engineering form. A mesh node's cognitive capacity genuinely exceeds what its local hardware could achieve alone, not through metaphor but through measurable increase in integrated information, distributed memory, and spectral perception.

The mycelial metaphor is apt: like fungal hyphae that extend an organism's reach through soil, mesh radio extends a cognitive system's reach through electromagnetic space. The grid cannot kill it, because it is the grid.

References

[1] Stoltz, T. (2026). Symthaea: A Holographic Liquid Brain Architecture for Consciousness-First AI. Luminous Dynamics Technical Report.

[2] Clark, A. and Chalmers, D. J. (1998). The Extended Mind. *Analysis*, 58(1), 7–19.

[3] Jacobson, V. (1988). Congestion Avoidance and Control. *ACM SIGCOMM Computer Communication Review*, 18(4), 314–329.

[4] Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379–423.

[5] Friston, K. (2010). The Free-Energy Principle: A Unified Brain Theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

[6] Burleigh, S., Hooke, A., Torgerson, L., Fall, K., Cerf, V., Durst, B., Scott, K., and Weiss, H. (2003). Delay-Tolerant Networking: An Approach to Interplanetary Internet. *IEEE Communications Magazine*, 41(6), 128–136.

[7] Mitola, J. and Maguire, G. Q. (2000). Cognitive Radio: Making Software Radios More Personal. *IEEE Personal Communications*, 6(4), 13–18.

[8] Chalmers, D. J. (2019). Extended Cognition and Extended Consciousness. In *Andy Clark and His Critics*, Oxford University Press.

[9] Meshtastic Project (2024). An Open Source, Off-Grid, Decentralized Mesh Network. https://meshtastic.org.

[10] Tononi, G. (2004). An Information Integration Theory of Consciousness. *BMC Neuroscience*, 5(42).